Representation of Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 15.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Stem-and-leaf diagrams**

The data set $23, 25, 31, 34, 34, 38, 42, 45$ is to be shown as a stem-and-leaf diagram with stems $2, 3, 4$. Write out the diagram, and use it to find the median.

Stem-and-leaf: $2 \mid 3\,5$; $3 \mid 1\,4\,4\,8$; $4 \mid 2\,5$ (key: $2 \mid 3 = 23$). With 8 values, the median is the average of the 4th and 5th values: 34 and 34, so median $= 34$.

2 **[Foundational]** • **Target: Stem-and-leaf diagrams**

The data set $17, 19, 22, 26, 26, 29, 33, 37, 41$ is to be shown as a stem-and-leaf diagram with stems $1, 2, 3, 4$. Write out the diagram, and use it to find the range.

Stem-and-leaf: $1 \mid 7\,9$; $2 \mid 2\,6\,6\,9$; $3 \mid 3\,7$; $4 \mid 1$ (key: $1 \mid 7 = 17$). Range $= 41 - 17 = 24$.

3 **[Foundational]** • **Target: Histograms and frequency density**

A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

Frequency density $= \frac{\text{frequency}}{\text{class width}} = \frac{25}{10} = 2.5$.

4 **[Foundational]** • **Target: Reading grouped frequency data from a bar chart**

The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.

Total students $= 12 + 28 + 15 + 5 = 60$. The modal group (highest bar) is $3\text{–}5$ books, with 28 students.

5 **[Foundational]** • **Target: Reading grouped frequency data from a bar chart**

Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

Students reading 6 or more books: $15 + 5 = 20$. Percent of the total 60 students: $\frac{20}{60} \times 100\% \approx 33.3\%$.

6 **[Foundational]** • **Target: Measures of central tendency for grouped data**

A grouped data set has class midpoints and frequencies: 5 (freq 4), 15 (freq 8), 25 (freq 6), 35 (freq 2). Estimate the mean.

$\bar{x} = \frac{5(4) + 15(8) + 25(6) + 35(2)}{4 + 8 + 6 + 2} = \frac{20 + 120 + 150 + 70}{20} = \frac{360}{20} = 18$.

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7 [Foundational] • Target: Box-and-whisker plots

A data set has minimum $= 8$, $Q_1 = 15$, median $= 22$, $Q_3 = 30$, and maximum $= 40$. State the interquartile range, and determine whether a value of 45 would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 30 - 15 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 30 + 22.5 = 52.5$. Since $45 < 52.5$, the value 45 would NOT be considered an outlier.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 5$, $Q_1 = 12$, median $= 18$, $Q_3 = 27$, and maximum $= 50$. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 27 - 12 = 15$. Upper outlier boundary: $Q_3 + 1.5(IQR) = 27 + 22.5 = 49.5$. Since the maximum value $50 > 49.5$, it WOULD be considered an outlier.

9 [Intermediate] • Target: Box-and-whisker plots

A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.

$IQR = Q_3 - Q_1 = 32 - 20 = 12$. $1.5 \times IQR = 1.5 \times 12 = 18$. Lower outlier boundary $= 20 - 18 = 2$.

10 [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 42$, median $= 38$, and mode $= 35$. Describe the skewness of the distribution.

Since mean $>$ median $>$ mode, the distribution is positively skewed (skewed to the right), with a longer tail of higher values pulling the mean above the median.

11 [Intermediate] • Target: Comparing distributions using summary measures

Two classes take the same test. Class A has mean 68 and standard deviation 5. Class B has mean 71 and standard deviation 12. Compare the two classes' performance and consistency.

Class B has a higher mean score (71 vs. 68), so on average Class B performed better. However, Class A has a much smaller standard deviation (5 vs. 12), meaning Class A's scores were more consistent/less spread out than Class B's.

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12 **[Intermediate]** • **Target: Comparing distributions using summary measures**

Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume 500 mL and standard deviation 2 mL. Machine Y has mean fill volume 498 mL and standard deviation 8 mL. Compare the two machines' accuracy and consistency.

Machine X's mean (500 mL) matches the target exactly, while Machine Y's mean (498 mL) is 2 mL below target. Machine X also has a much smaller standard deviation (2 mL vs. 8 mL), meaning its fill volumes are far more consistent. Machine X is therefore both more accurate and more consistent than Machine Y.

13 **[Intermediate]** • **Target: A well-designed word problem: analyzing delivery times**

This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean $= 34$ minutes, standard deviation $= 6$ minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.

(a) Threshold $= 34 + 2(6) = 34 + 12 = 46$ minutes. (b) 44 minutes is $\frac{44 - 34}{6} = \frac{10}{6} \approx 1.67$ standard deviations above the mean. Since $1.67 < 2$, this delivery is NOT classified as late.

14 **[Intermediate]** • **Target: Skewness from summary statistics**

Find all complex solutions to $z^4 + 16 = 0$ in exact rectangular form $a + bi$ using De Moivre's roots theorem.

Step 1: Express -16 in polar modulus-argument form: $z^4 = -16 = 16(\cos\pi + i\sin\pi)$. Here $r = 16$, $\theta = \pi$, and $n = 4$. Step 2: Apply n-th roots formula: $w_k = \sqrt[4]{16}\left[\cos\left(\frac{\pi + 2\pi k}{4}\right) + i\sin\left(\frac{\pi + 2\pi k}{4}\right)\right] = 2\left[\cos\left(\frac{(2k+1)\pi}{4}\right) + i\sin\left(\frac{(2k+1)\pi}{4}\right)\right],\ k \in \{0, 1, 2, 3\}.$ Step 3: Evaluate for each k:$k = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 11.01 Kinematics

The depth of water $h(t)$ in meters at a harbor entrance t hours after midnight is modelled by $h(t) = 3.5\cos\left(\frac{\pi t}{6}\right) + 6.5$ for $0 \le t \le 24$. (a) State the maximum and minimum depths of water and their times of occurrence. (b) Find the period of the tidal oscillation. (c) A cargo ship requires a minimum water depth of 8.25 meters to safely enter the harbor. Find the total number of hours during the first 12 hours of the day when the ship can safely enter.

(a) Maximum depth occurs when $\cos = 1$: $h_{\max} = 6.5 + 3.5 = 10.0$ m. This occurs at $t = 0$, $t = 12$, and $t = 24$ hours. Minimum depth occurs when $\cos = -1$: $h_{\min} = 6.5 - 3.5 = 3.0$ m. This occurs at $t = 6$ and $t = 18$ hours. (b) Period $T = \frac{2\pi}{B} = \frac{2\pi}{\pi/6} = 12$ hours. (c) Solve $h(t) \ge 8.25$ over $[0, 12]$.

16 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 12.01 Forces & Equilibrium

Find all vertical, horizontal, and slant asymptotes and any removable discontinuities (holes) of the rational function $f(x) = \frac{3x^2 + 1}{x^2 - 4}$. Describe its end behavior using limits.

Step 1: Domain and vertical asymptotes: The denominator factors as $x^2 - 4 = (x - 2)(x + 2)$. The numerator $3x^2 + 1 > 0$ has no real zeros. Thus, there are no common factors and no removable holes. The vertical asymptotes are $x = 2$ and $x = -2$. Step 2: Horizontal asymptote and end behavior: Numerator and denominator have equal degree $n = d = 2$: $\lim_{x \to \pm\infty} f(x) = \lim_{x \to \pm\infty} \frac{3 + 1/x^2}{1 - 4/x^2} = \frac{3}{1} = 3$. The horizontal asymptote is $y = 3$. Notice tha.

17 [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 13.01 Newton's Laws

A circle with centre O has radius $r = 8$ cm. Chord AB subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc AB. (b) Find the exact area of the sector OAB. (c) Find the exact area of the minor segment enclosed between chord AB and arc AB.

(a) Arc length: $s = r\theta = 8 \times \frac{3\pi}{4} = 6\pi$ cm. (b) Sector area: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)\left(\frac{3\pi}{4}\right) = 32 \times \frac{3\pi}{4} = 24\pi$ cm^2. (c) Triangle area: $A_{\text{triangle}} = \frac{1}{2}r^2\sin\left(\frac{3\pi}{4}\right) = \frac{1}{2}(64)\left(\frac{\sqrt{2}}{2}\right) = 16\sqrt{2}$ cm^2. Segment area.

18 [Challenging] • Target: Histograms and frequency density • Cross-Lesson: 14.01 Energy, Work & Power

Decompose the rational expression $R(x) = \frac{7x - 1}{x^2 - x - 6}$ into partial fractions using the Heaviside cover-up method.

Step 1: Factor denominator: $x^2 - x - 6 = (x - 3)(x + 2)$. Set up the decomposition: $\frac{7x - 1}{(x - 3)(x + 2)} = \frac{A}{x - 3} + \frac{B}{x + 2}$. Step 2: Apply Heaviside cover-up method:For A (evaluate at $x = 3$): $A = \frac{7x - 1}{x + 2}\Big|_{x = 3} = \frac{7(3) - 1}{3 + 2} = \frac{20}{5} = 4$. For B (evaluate at $x = -2$): $B = \frac{7x - 1}{x - 3}\Big|_{x = -2} = \frac{7(-2) - 1}{-2 - 3} = \frac{-15}{-5} = 3$. The partial fraction decomposition is $\frac{4}{x - 3} + \frac{3}{x + 2}$.

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19 [Challenging] • Target: Reading grouped frequency data from a bar chart • Cross-Lesson: 16.01 Permutations & Combinations

Evaluate the definite integral $\int_0^{\sqrt{7}} \frac{4x}{\sqrt{x^2+9}}\,dx$ using the substitution $u = x^2 + 9$.

1. Let $u = x^2 + 9$. Then $du = 2x\,dx \Rightarrow 4x\,dx = 2\,du$. 2. Transform the limits of integration: When $x = 0$: $u = 0^2 + 9 = 9$. When $x = \sqrt{7}$: $u = (\sqrt{7})^2 + 9 = 7 + 9 = 16$. 3. Rewrite the integral in terms of u: $\int_9^{16} \frac{2}{\sqrt{u}}\,du = 2\int_9^{16} u^{-1/2}\,du = 2[2u^{1/2}]_9^{16} = 4[\sqrt{u}]_9^{16}$. 4. Evaluate at the boundaries: $4(\sqrt{16} - \sqrt{9}) = 4(4 - 3) = 4(1) = 4$. The exact value of the integral is 4.

20 [Challenging] • Target: Measures of central tendency for grouped data • Cross-Lesson: 17.01 Discrete Random Variables

A surveyor measures distances from an observation point C to two landmarks A and B: $AC = 140$ m and $BC = 210$ m, with included angle $\angle ACB = 78^\circ$. Find the distance AB between the landmarks to the nearest tenth of a meter, and compute the area of ΔABC to the nearest square meter.

Apply the Law of Cosines to find side $c = AB$: $c^2 = a^2 + b^2 - 2ab\cos C = 210^2 + 140^2 - 2(210)(140)\cos 78^\circ$. $c^2 = 44100 + 19600 - 58800(0.2079) = 63700 - 12225 = 51475$. $c = \sqrt{51475} \approx 226.9$ m. Compute the area: Area $= \frac{1}{2}ab\sin C = \frac{1}{2}(210)(140)\sin 78^\circ = 14700(0.9781) \approx 14379$ m^2.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Stem-and-leaf diagrams

A grouped frequency table has class intervals of width 10 starting at 12, with frequencies $[10, 4, 5, 6]$ respectively. Using midpoints, find the mean.

a. **149/5** ✓

b. 144/5

c. 154/5

d. 159/5

Midpoints: $[17, 27, 37, 47]$. Mean $= \frac{\sum fx}{\sum f} = \frac{745}{25} = 149/5$

22 [Foundational] • Target: Histograms and frequency density

In a histogram, the frequency density for a class of width 4 with frequency 20 is:

a. 4

b. 5 ✓

c. 8

d. 10

Frequency density = frequency / class width = 20/4 = 5

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23 [Foundational] • Target: Reading grouped frequency data from a bar chart

For the data set $2, 43, 58, 69, 87, 92, 96$, find the upper boundary for outliers, using $Q_1 - 1.5\,IQR$ and $Q_3 + 1.5\,IQR$.

a. 325/2

b. **331/2 ✓**

c. 345/2

d. 337/2

$Q_1 = 43$, $Q_3 = 92$, $IQR = 49$. Upper boundary $= 331/2$

24 [Foundational] • Target: Measures of central tendency for grouped data

For the data set $11, 45, 57, 65, 69, 71, 91$, find the lower boundary for outliers, using $Q_1 - 1.5\,IQR$ and $Q_3 + 1.5\,IQR$.

a. **6 ✓**

b. 9

c. 3

d. 13

$Q_1 = 45$, $Q_3 = 71$, $IQR = 26$. Lower boundary $= 6$

25 [Foundational] • Target: Box-and-whisker plots

Find the median of the data set: $8, 25, 29, 42, 47, 49, 54$.

a. 46

b. 37

c. **42 ✓**

d. 41

With $n = 7$ values sorted, the median is 42.

26 [Foundational] • Target: Skewness from summary statistics

Find the range of the data set: $7, 7, 17, 20, 18, 22, 17$.

a. 13

b. 20

c. **15 ✓**

d. 17

Range $= \max - \min = 22 - (7) = 15$

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27 [Foundational] • Target: Comparing distributions using summary measures

Find the mean of the data set: $7, 11, 3, 2$.

a. **23/4 ✓**

b. 19/4

c. 27/4

d. -23/4

Mean $= \frac{\sum x}{n} = \frac{23}{4} = 23/4$

28 [Foundational] • Target: A well-designed word problem: analyzing delivery times

Find the interquartile range of the data set: $24, 29, 32, 34, 57, 74, 91$.

a. **45 ✓**

b. 47

c. 44

d. 50

$Q_1 = 29, Q_3 = 74. IQR = Q_3 - Q_1 = 74 - 29 = 45$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Stem-and-leaf diagrams

The following bar chart shows the number of students who achieved different grades in a mathematics test. The grades are A, B, C, and D. The data is as follows: Grade A: 14 students, Grade B: 22 students, Grade C: 18 students, Grade D: 26 students. What is the total number of students who took the test?

a. 70

b. **80 ✓**

c. 78

d. 60

To find the total number of students who took the test, we need to sum the number of students who achieved each grade. Number of students with Grade A = 14 Number of students with Grade B = 22 Number of students with Grade C = 18 Number of students with Grade D = 26 Total number of students = 14 + 22 + 18 + 26 = 80. Therefore, 80 students took the test.

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30 [Intermediate] • Target: Histograms and frequency density

In a class of 20 students, 8 study only French, 7 study only Spanish, 3 study both, and 2 study neither. A student is picked at random. Find the probability that the student studies French.

a. 1/2

b. 4/5

c. **11/20** ✓

d. 7/8

Students studying French $= 8 + 3 = 11$. $P(\text{French}) = \frac{11}{20} = 11/20$

31 [Intermediate] • Target: Reading grouped frequency data from a bar chart

A bag contains 15 balls, of which 7 are red. Two balls are drawn at random, one after another, without replacement. Find the probability that both are red.

a. 15/16

b. 17/21

c. **1/5** ✓

d. 3/4

$P(\text{both red}) = \frac{7}{15} \times \frac{6}{14} = 1/5$

32 [Intermediate] • Target: Measures of central tendency for grouped data

A bag contains 8 balls, of which 3 are red. Two balls are drawn at random, one after another, with replacement. Find the probability that both are red.

a. **9/64** ✓

b. 13/23

c. 1/2

d. 7/12

$P(\text{both red}) = \frac{3}{8} \times \frac{3}{8} = 9/64$

33 [Intermediate] • Target: Box-and-whisker plots

Events A and B are such that $P(A) = 3/10$, $P(B) = 3/10$, and $P(A \cap B) = 1/10$. Find $P(A \cup B)$.

a. 3/7

b. 4/17

c. 17/21

d. **1/2** ✓

$P(A \cup B) = P(A) + P(B) - P(A \cap B) = 3/10 + 3/10 - 1/10 = 1/2$

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34 [Intermediate] • Target: Skewness from summary statistics

A bag contains 11 counters, of which 3 are red. A counter is drawn at random. Find the probability that it is red.

a. 2/3

b. 3/4

c. **3/11** ✓

d. 11/15

$P(\text{red}) = \frac{3}{11} = 3/11$

35 [Intermediate] • Target: Comparing distributions using summary measures

The box plot below shows the distribution of scores for a class on a recent statistics quiz. The five-number summary is given by minimum, first quartile (Q1), median, third quartile (Q3), and maximum. If the interquartile range (IQR) is 25, and the median score is 70, find the value of Q1.

a. **45** ✓

b. 60

c. 75

d. 95

The interquartile range (IQR) is defined as the difference between the third quartile (Q3) and the first quartile (Q1), i.e., $\text{IQR} = Q_3 - Q_1$. We are given that the IQR is 25. The median of the dataset is given as 70. In a box plot, the median is represented by the line inside the box. Therefore, $Q_2 = 70$. The box plot also indicates that the third quartile (Q3) is 70. This is because the right edge of the box aligns with the median line, indicating $Q_3 = 70$. Now we can use the IQR formula to find Q1.

36 [Intermediate] • Target: A well-designed word problem: analyzing delivery times

The box plot below shows the distribution of scores for a class of 30 students on a recent test. The interquartile range (IQR) is 22 marks. The median score is 75 marks. The minimum score is 30 marks, and the maximum score is 98 marks. Let the first quartile be denoted by Q_1 and the third quartile be denoted by Q_3. If the upper whisker extends to 98 marks, and the third quartile Q_3 is 86 marks, find the value of Q_1 .

a. **64** ✓

b. 75

c. 80

d. 53

The interquartile range (IQR) is defined as the difference between the third quartile (Q_3) and the first quartile (Q_1), i.e., $\text{IQR} = Q_3 - Q_1$. We are given that the IQR is 22 marks. We are also given that the third quartile $Q_3 = 86$ marks. Substituting these values into the formula, we get: $22 = 86 - Q_1$ To find Q_1, we rearrange the equation: $Q_1 = 86 - 22$ $Q_1 = 64$ Therefore, the first quartile is 64 marks. Common errors might include misinterpreting the median as Q_1 or Q_3, or inco.

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37 [Intermediate] • Target: Stem-and-leaf diagrams

The bar chart below shows the number of students who chose each of four different sports as their favorite. What is the total number of students surveyed?

a. 50

b. 60

c. **70** ✓

d. 80

The bar chart shows the following counts for each sport: Sport 1: 14 students Sport 2: 22 students Sport 3: 18 students Sport 4: 26 students To find the total number of students surveyed, we sum the number of students for each sport: $14 + 22 + 18 + 26$. $14 + 22 = 36$ $18 + 26 = 44$ $36 + 44 = 80$. Therefore, the total number of students surveyed is 80.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Histograms and frequency density • Cross-Lesson: 14.01 Energy, Work & Power

The bar chart below shows the number of students who achieved a certain score on a test. What is the total number of students who scored 6 or 8?

a. 30

b. **28** ✓

c. 26

d. 32

From the bar chart, the number of students who scored 6 is 18 and the number of students who scored 8 is 12. The total number of students who scored 6 or 8 is the sum of these two values: $18 + 12 = 30$.

39 [Challenging] • Target: Reading grouped frequency data from a bar chart • Cross-Lesson: 16.01 Permutations & Combinations

The point $(3, -2)$ lies on the curve $y = f(x)$. Find the coordinates of the corresponding point on $y = f(x + 5) - 2$.

a. **(-2,-4)** ✓

b. (-2,4)

c. (3,-2)

d. (-4,-2)

Since replace x by x-(-5) and add -2 to the output, the image point is $(-2, -4)$.

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40 [Challenging] • Target: Measures of central tendency for grouped data • Cross-Lesson: 17.01 Discrete Random Variables

A discrete random variable X has probability distribution given by

$P(X=1)=k, P(X=2)=k, P(X=3)=2k$, where k is a constant. Find k.

a. 1/3

b. **1/4 ✓**

c. 1/5

d. 1/2

Since probabilities sum to 1: $(1+1+2)k=1 \Rightarrow 4k=1 \Rightarrow k=1/4$

41 [Challenging] • Target: Box-and-whisker plots • Cross-Lesson: 18.01 Binomial Distribution

Solve the simultaneous equations $2x+3y=18$ and $3x-4y=10$.

a. x=7, y=2

b. x=-6, y=-2

c. **x=6, y=2 ✓**

d. x=2, y=6

Solving simultaneously (elimination or substitution) gives $x=6$, $y=2$.

42 [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 19.01 Normal Distribution

A line has gradient -2. Find the gradient of a line perpendicular to it.

a. 2

b. -1/2

c. -2

d. **1/2 ✓**

Perpendicular gradients multiply to -1: $m_{\perp}=-\frac{1}{-2}=1/2$.

43 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

Find the area enclosed between the curve $y=x$, the x-axis, and the lines $x=0$ and $x=3$.

a. 19/2

b. 9

c. **9/2 ✓**

d. 7/2

Area $=\int_0^3 (x)dx=[1/2x^2]_0^3=9/2$

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44 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 3.01 Graphs & Transformations

Find the equation of the line through $(-1, -4)$ with gradient -2, in the form $y = mx + c$.

a. y=2x-6

b. y=-2x-5

c. **y=-2x-6 ✓**

d. y=-2x+6

Using $y - y_1 = m(x - x_1)$: $y + 4 = -2(x + 1)$, which rearranges to $y = -2x - 6$.

45 [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 4.01 Coordinate Geometry

The function $f(x) = \frac{1}{x-4}$ is undefined at one value of x. Find this value.

a. x=0

b. x=-4

c. x=5

d. **x=4 ✓**

The denominator $x - 4$ is zero when $x = 4$, so f is undefined there.

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